

Muhammad Maaz Aslam Sonaly

AI Engineer · Forward Deployed Engineer · Full-Stack & Platform

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SUMMARY

AI engineer who builds applied-AI products and the production systems that run them. At YoC I architected a DB-coordinated distributed call-analytics pipeline on Google Gemini, fine-tuned domain-specific LLM chatbots (LoRA/QLoRA/PEFT) grounded with RAG, and designed and **solely operate** a self-hosted Kubernetes (k3s + GitOps) platform hosting ~13 production apps — end to end, from architecture to deployment. Comfortable across the full stack (React/TypeScript, Node/Express, Python, PostgreSQL) with a strong DevOps/cloud backbone. Winner of the **2026 YoC Emerging Talent award (AI Automation)**. Computer & Information Systems Engineering graduate, NED University.

TECHNICAL SKILLS

AI & Machine Learning: LLM fine-tuning (LoRA, QLoRA, PEFT), Retrieval-Augmented Generation (RAG), Chroma / pgvector, Google Gemini (multimodal), Hugging Face Transformers, wav2vec2 / XLSR-53, CTC, mT5, PyTorch, embeddings, tokenization, quantization, inference optimization, Whisper, EasyOCR

Languages: TypeScript, JavaScript, Python, PHP, SQL, C#, Dart, C

Frontend: React 19, Vite, Tailwind CSS, TanStack Query, Recharts, Flutter

Backend: Node.js, Express, NestJS, Prisma, Laravel, .NET 8, REST API design, WebSockets, OpenAPI / Zod

Databases: PostgreSQL, Supabase, MySQL, Google BigQuery, pgvector

DevOps & Cloud: Kubernetes (k3s), Argo CD (GitOps), Docker (multi-stage), GitHub Actions (CI/CD), GitHub Container Registry, Traefik, cert-manager + Let's Encrypt, Google Cloud (Compute Engine, Cloud Storage, Pub/Sub), Cloudflare, AWS S3, Linux, Nginx

Security & Integrations: Google OAuth 2.0 / OpenID Connect, JWT, RBAC, PostgreSQL row-level security, email deliverability (SPF / DKIM / DMARC), PayPro payments, Google Drive & Business Profile APIs, webhooks

EXPERIENCE

AI Engineer → Forward Deployed Engineer — YoC (Your Online Conversation)

Full-time · Remote / United Kingdom · Jul 2025 – Present

Sole engineer for a UK travel business — owning its AI systems, product suite, and delivery platform from architecture through deployment and operations.

- Architected a **DB-coordinated distributed worker system** that processes **500+ hour-long call recordings in ~10 minutes** — dynamically scalable workers with adjustable throughput, batched polling, and parallel async execution — integrating **Google Gemini (multimodal)** for high-throughput call analysis.
- Built and fine-tuned domain-specific **LLM chatbots with LoRA, QLoRA, and PEFT**, grounded by **RAG over a Chroma vector database**; built tokenization, embedding, quantization, and inference-optimization pipelines on Hugging Face Transformers.
- Designed, built, and **solely operate a self-healing GitOps Kubernetes platform** (k3s + Argo CD) on a hardened Google Cloud VM, migrating **~13 production apps off shared cPanel** onto a single cloud server — deployments became a git push with auto-sync, self-healing, per-app isolation, Traefik ingress, and automatic TLS (cert-manager + Let's Encrypt).
- Built a **multi-source airline accounting reconciliation and fraud-detection system** that normalizes 8 GDS/airline report formats into a canonical ticket key and flags variances **to the penny (£0.01)**, with a daily-totals engine and an admin fraud-review workflow.
- Delivered **AI-driven automation** for CRM upselling, accounts reconciliation, and variance detection; shipped SaaS products end to end from architecture through deployment and scaling.
- Automated **AI marketing-email generation**, cutting HTML template complexity from 2,000+ lines to ~200, plus production email deliverability (DMARC, one-click unsubscribe, throttled bulk sending).
- Built a **cross-app data lake** (Python ETL → Google Cloud Storage Parquet / BigQuery with schema auto-discovery and validation) and **single sign-on** across the suite (Google OAuth 2.0 / OIDC, JWTs, cross-subdomain RBAC).
- Engineered **CI/CD with GitHub Actions** and custom webhooks for Node.js and Python services, and automated data pipelines for web scraping, cleaning, preprocessing, and ML dataset structuring.
- **Recognition:** YoC *Emerging Talent* award, AI Automation Department (Annual Award Ceremony, May 2026).

Information Technology Intern — Wolke Consultancy

Karachi, Pakistan · On-site · Apr 2025 – Jul 2025

- Delivered custom **WordPress** sites for international clients, scoping directly with US-based stakeholders; handled plugin troubleshooting, **SEO optimization**, and multi-environment development.
- Built a pet-daycare application with web and mobile front-ends using **.NET 8** (API, routing, MVC) and **React**.
- Customized and performance-tuned sites with PHP, HTML, and CSS.

Freelance Web Developer — Upwork

Remote · Dec 2023 — May 2025

- Delivered web development for international clients over an 18-month engagement, building responsive sites with HTML, CSS, and PHP; owned client communication from scoping through delivery.

Web Development Intern — Fossphorus

On-site · Nov 2023 — Jan 2024

- Converted static pages into dynamic **Laravel (PHP)** applications, moving logic server-side to cut redundant markup and calls; performed testing, debugging, and QA with senior developers.

SELECTED PROJECTS

Memoni Low-Resource Language AI (Research) · PyTorch, Hugging Face Transformers, wav2vec2/XLSR-53, mT5

Speech recognition and Urdu→Memoni translation for an unwritten language: a cross-script transliteration-folding pipeline, a pivot-language pseudo-label bootstrap (Whisper → mT5) to create training data where none existed, and a fine-tuned wav2vec2/CTC model — deployed with a community labeling app on Hugging Face Spaces.

OmiQo — Marketplace App · Flutter, NestJS, Prisma, PostgreSQL, Socket.io, AWS S3

Solo-built customer and vendor mobile apps with real-time Socket.io order queues, PayPro payments (tokenized auth, webhook reconciliation, retry logic), an S3 document pipeline, and unified JWT access/refresh auth across both apps.

RL-Based Dynamic Job Scheduling (Final Year Project) · Reinforcement Learning, Python

Designed a reinforcement-learning scheduler for deadline-aware dispatch — full MDP formulation, ETA-accuracy reward design, and algorithm analysis (DQN / PPO / A2C); inspired by Uber's DeepETA.

Hotel Management System (Laravel, Livewire, MySQL) — booking platform with admin approval workflow and SMTP confirmations. · **Car Data Analysis** (Python, pandas) — vehicle-telematics pipeline with trip/speed/acceleration statistics and anomaly detection.

EDUCATION

B.E. in Computer & Information Systems Engineering

NED University of Engineering & Technology, Karachi · Sep 2022 — 2026

Relevant coursework: Data Structures & Algorithms, Database Systems, Operating Systems, Computer Networks, Machine Learning, Artificial Intelligence, Software Engineering, Computer Systems Security.

CERTIFICATIONS & AWARDS

- YoC Emerging Talent Award — AI Automation Department, 2026
- Machine Learning Specialization — Stanford University (Coursera), 2025
- Google Data Analytics Professional Certificate — Google, 2024
- Introducing Generative AI with AWS — Udacity, 2024 · Azure AI Fundamentals — Microsoft Learn, 2024